MAM200W Module 2LA: Linear Algebra

Chapter 3 **Some answers to the exercises** **2023**

Section 3.1

1. (c) Let $V = W = \mathbb{R}$ and $F(x) = x^2$.

2. (a) Yes (b) Yes (c) No (d) Yes (e) No (f) No (g) Yes
   The proofs for (b) and (c) are given below.
   (b) For all real numbers $a_0, a_1, a_2, b_0, b_1, b_2, k$ we have:

$$\begin{aligned} & T((a_0 + a_1x + a_2x^2) + (b_0 + b_1x + b_2x^2)) \\ = \; & T((a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2) \\ = \; & (a_0 + b_0) + (a_1 + b_1)(x + 1) + (a_2 + b_2)(x + 1)^2 \\ = \; & a_0 + a_1(x + 1) + a_2(x + 1)^2 + b_0 + b_1(x + 1) + b_2(x + 1)^2 \\ = \; & T(a_0 + a_1x + a_2x^2) + T(b_0 + b_1x + b_2x^2) \end{aligned}$$

   So T preserves addition. Also:

$$\begin{aligned} & T(k(a_0 + a_1x + a_2x^2)) \\ = \; & T(ka_0 + ka_1x + ka_2x^2) \\ = \; & ka_0 + ka_1(x + 1) + ka_2(x + 1)^2 \\ = \; & k(a_0 + a_1(x + 1) + a_2(x + 1)^2) \\ = \; & kT(a_0 + a_1x + a_2x^2) \end{aligned}$$

   So T also preserves scalar multiplication.
   (c) Since $L(0+0x+0x^2) = 1+x+x^2$, L does not map the zero vector to the zero vector, so L is not a linear transformation. (See Corollary 3.1.9)

3. Suppose that $\mathcal{F}f_1$ and $\mathcal{F}f_2$ are Fourier transforms of f_1 and f_2 respectively. We now show that $\mathcal{F}f_1 + \mathcal{F}f_2$ is a Fourier transform of $f_1 + f_2$ (i.e. that $T(f_1) + T(f_2) = T(f_1 + f_2)$).
   So we show that $(\mathcal{F}f_1+\mathcal{F}f_2)(w) = (\mathcal{F}(f_1+f_2))(w)$ for all w, as follows:

$$\begin{aligned} (\mathcal{F}f_1 + \mathcal{F}f_2)(w) &= \mathcal{F}f_1(w) + \mathcal{F}f_2(w) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iwt} f_1(t)\, dt + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iwt} f_2(t)\, dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iwt} (f_1(t) + f_2(t))\, dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iwt} (f_1 + f_2)(t)\, dt \\ &= (\mathcal{F}(f_1 + f_2))(w) \end{aligned}$$

Next we show that T preserves scalar multiplication. We check that if $\mathcal{F}f$ is a Fourier transform of f, then $k\,\mathcal{F}f$ is a Fourier transform of kf, for any scalar k. For all w:

$$\begin{aligned}
k(\mathcal{F}f(w)) &= k\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-iwt} f(t)\,dt \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-iwt} k\,f(t)\,dt \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-iwt} (k\,f)(t)\,dt \\
&= (\mathcal{F}(k\,f))(w).
\end{aligned}$$

4. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation. We show that T maps a line in $\mathbb{R}^3$ to a line (or a point) in $\mathbb{R}^3$. Let's begin with the line with vector equation $(x, y, z) = (a_1, b_1, c_1) + t(a_2, b_2, c_2),\ t \in \mathbb{R}$. The set of all points on this line is

$$\{(a_1, b_1, c_1) + t(a_2, b_2, c_2)\ :\ t \in \mathbb{R}\}.$$

Now apply T to each point in this set to get the new set

$$\{T(a_1, b_1, c_1) + t\,T(a_2, b_2, c_2)\ :\ t \in \mathbb{R}\}.$$

Note that we are using linearity of T here. This new set of points again forms a line (or a point, if the vector $T(a_2, b_2, c_2) = (0, 0, 0)$).

Section 3.2

1. (a) $\ker(T) = \left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$ and $\ker(S) = \left\{t\begin{pmatrix} 1 \\ 1 \end{pmatrix}\ :\ t \in \mathbb{R}\right\}$

   (b) $\operatorname{ran}(T) = \mathbb{R}^2$ and $\operatorname{ran}(S) = \left\{t\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\ :\ t \in \mathbb{R}\right\}$

2. (b) only

3. (a) only

4. (a) $\ker(T) = \{\vec{0}\}$ (b) $\operatorname{ran}(T) = V$

5. (a) Yes (b) 1

6. (a) $\left\{\begin{pmatrix} 3 \\ -8 \\ 4 \\ 3 \end{pmatrix}\right\}$ (b) $\left\{\begin{pmatrix} 1 \\ 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \\ -1 \end{pmatrix}\right\}$

7. (a) No (b) Yes (c) Yes (d) No

   (e)

$$\begin{aligned}
\ker(L) &= \{at^2 + bt + c \,:\, (a+c)t^2 + (b+c)t = 0.t^2 + 0.t + 0 \text{ and } a, b, c \in \mathbb{R}\} \\
&= \{at^2 + bt + c \,:\, a + c = b + c = 0 \text{ and } a, b, c \in \mathbb{R}\} \\
&= \{at^2 + bt + c \,:\, a = -c = b \text{ and } a, b, c \in \mathbb{R}\} \\
&= \{-ct^2 - ct + c \,:\, c \in \mathbb{R}\} \\
&= \{c(-t^2 - t + 1) \,:\, c \in \mathbb{R}\}
\end{aligned}$$

   So a basis for $\ker(L)$ is $\{-t^2 - t + 1\}$.
   (f) We apply Theorem 3.2.10, using the standard basis $S = \{1, t, t^2\}$ for P_2. So $L^{\rightarrow}(S) = \{t^2 + t, t, t^2\}$. This set spans $\mathrm{ran}(L)$, but is dependent, because $t^2 + t$ is a linear combination of the other two vectors in the set. We eliminate it, and are left with $\{t, t^2\}$ as a basis for $\mathrm{ran}(L)$. [The idea was that you'd get a bit of practice using Theorem 3.2.10, but admittedly it's easy to do this just using the definition of range.]

8. (a) $\{1\}$ (b) $\{t^2, t, 1\}$

9. (a) $(1, -1)$
   (c) $\ker(T) = \{p(x) \in P_1 \,:\, p(0) = 0 \text{ and } p(1) = 0\}$. Using the form $p(x) = ax+b$ for members of P_1 gives $\ker(T) = \{ax+b \,:\, a = 0 \text{ and } b = 0\} = \{\vec{0}\}$. So T is one-one.
   (d) Yes. Check that it is surjective.

10. Rot_x and Ref_x are isomorphisms.

11. (a) If S is linearly dependent, then one of the vectors in S can be written as a linear combination of the others. Then the image of this vector under T is a linear combination of the images of the others.
    (b) The zero transformation will do.

12. Check that putting $T(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n) = a_1\mathbf{w}_1 + \cdots + c_n\mathbf{w}_n$ will do the trick.

Section 3.3

2(b) It is enough to check that F is injective and surjective. The first follows from the fact that a basis is linearly independent, the second from the fact that if $(\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^n$, then $\lambda_1\mathbf{b}_1 + \cdots + \lambda_n\mathbf{b}_n \in V$, where $B = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$.
$F^{-1}((\lambda_1, \ldots, \lambda_n)) = \lambda_1\mathbf{b}_1 + \cdots + \lambda_n\mathbf{b}_n$.

3(a) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & -2 & -3 \end{pmatrix}$

4. $\begin{pmatrix} 0 & 0 \\ -\frac{1}{2} & 1 \\ \frac{8}{3} & \frac{4}{3} \end{pmatrix}$

5. The matrix is an $(n+1)\times(n+1)$ matrix with first column containing only zeros and column k having zeros everywhere except in position $k-1$, where it has a k.

6(a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 1 \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{3}{2} \end{pmatrix}$

7(a) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 1 & 0 & -1 \\ -1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$

9. $\begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$